

EFFECTS OF TWO TRAINING PROTOCOLS ON THE FOREHAND DRIVE PERFORMANCE IN TENNIS

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ABSTRACT

Genevois, C, Frican, B, Creveaux, T, Hautier, C, and Rogowski, I. Effects of two training protocols on the forehand drive performance in tennis. *J Strength Cond Res* 27(3): 677–682, 2013—The aim of this study was to examine the effects of 2 training modalities on the tennis forehand drive performance. Forty-four tennis players (mean $\pm$ SD: age = 26.9 $\pm$ 7.5 years; height = 178.6 $\pm$ 6.7 cm; mass = 72.5 $\pm$ 8.0 kg; International Tennis Number = 3) were randomly assigned into 3 groups. During 6 weeks, the first group performed handled medicine ball (HMB) throws included in the regular tennis practice, the second group (overweight racket—OWR) played tennis forehand drives with an overweighed racket during the regular tennis practice, and the third group (regular tennis training—RTT) practiced only tennis training as usual. Before and after the 6-week program, velocity and accuracy of tennis crosscourt forehand drives were evaluated in the 3 groups. The main results showed that after 6-week training, the maximal ball velocity was significantly increased in HMB and OWR groups in comparison with RTT ($p < 0.001$ and $p = 0.001$, respectively). The estimated averaged increase in ball velocity was greater in HMB than in OWR (11 vs. 5%, respectively; $p = 0.017$), but shot accuracy tended to be deteriorated in HMB when compared with OWR and RTT ($p = 0.043$ and $p = 0.027$, respectively). The findings of this study highlighted the efficiency of both training modalities to improve tennis forehand drive performance but also suggested that the HMB throws may be incorporated into the preseason program preferably, whereas the OWR forehand drives may be included in the on-season program.

KEY WORDS special strength training, specific strength training, handled medicine ball, overweight racket, ground stroke

INTRODUCTION

One of the characteristics for the tennis evolution over the past decade is a preferential use of the forehand drive in the construction of the point (15) that appears as a key stroke of the modern game (2). Therefore, the improvement of the drive speed, while maintaining a great level of control, is one of the factors of performance to succeed in tennis. As consequence, strength training has become vital in the modern game as it contributes to increase in ball velocity.

The ultimate goal of resistance training is typically to increase muscular power under the variety of conditions that relate to tennis performance (17). To develop the quality of power, several authors recommend using explosive exercises, such as medicine ball throws (1,3,17,18,23), but to the best of our knowledge, no experimental data are available to demonstrate the efficiency of such method for tennis ground stroke improvement. Another way to develop the quality of power is to provide a training stimulus that mimics motions and bioenergetics systems involved in the activity using overweight implements. Several data underline the efficiency of this training modality, such as in baseball (13) or soccer (22). Although Cardoso Marques (3) suggests such training contents for tennis player, no experimental data confirm the efficiency of overweight racket (OWR) training modality based on real shots to enhance tennis ground stroke performance.

Thus, this study aimed at testing the efficiency of 2 training modalities, one based on handled medicine ball (HMB) throws and the other based on overweight racket (OWR), on the forehand drive performance. It was hypothesized that HMB throws would allow a higher increase in forehand drive velocity than playing with OWR and that the difference between throwing and forehand drive technique would lead to an alteration in the shot accuracy.

METHODS

Experimental Approach to the Problem

Forty-four tennis players were randomly assigned into 3 groups. During the 6-week training, the first group added HMB throws in its training (HMB group), the second group included forehand drives played with an OWR (OWR

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TABLE 1. Characteristics of the 3 groups of players, with HMB throws modality, OWR training modality, and RTT group.*

	HMB	OWR	RTT
<i>n</i>	12	20	12
Age (y)	27.6 ± 6.1	27.0 ± 7.8	26.6 ± 8.6
Height (cm)	179.5 ± 5.9	178.9 ± 7.4	177.4 ± 6.7
Mass (kg)	75.0 ± 7.6	72.0 ± 8.3	72.1 ± 8.8
Current ranking	13 ± 2	12 ± 3	13 ± 3
Tennis practice (y)	18.8 ± 6.1	17.2 ± 7.0	14.3 ± 3.8
Weekly tennis training (h)	2.0 ± 0.9	4.4 ± 3.1	3.0 ± 1.0
Weekly conditioning (h)	1.5 ± 0.8	2.0 ± 1.4	1.8 ± 1.8

*HMB = handled medicine ball; OWR = overweight racket; RTT = regular tennis training.

group), and the third group made regular tennis training (RTT group). The study was a randomized controlled trial. All training sessions were supervised by 2 tennis physical trainers, the 2 lead authors. Velocity and accuracy of tennis forehand drives were tested in the 3 groups before and after the 6-week program.

Subjects

Forty-four male tennis players (mean ± SD: age = 26.9 ± 7.5 years; height = 178.6 ± 6.7 cm; mass = 72.5 ± 8.0 kg; International Tennis Number = 3) gave their written informed consent to participate in this study, which was approved by the French Ethics Committee Sud Est II. None of them suffered injury during the 6 months before the study. The procedure of the experiment and tasks were explained, whereas no information was provided about the expected results of the study. The characteristics of the groups are presented in Table 1.

Procedures

All players were tested with their personal racket before and after a 6-week training protocol. Pre- and post-tests were conducted inside on green set court with the same testing

procedures and format. After a general and specific warm-up, which included forehand drives, players were instructed to hit 2 sets of 10 crosscourt forehand drives as fast as possible in a target zone. New tennis balls were projected by a ball machine (Airmatic 104; Pop-lob, Paris, France) without spin and in the same area requiring the players only a small stance adjustment to hit the ball. The machine was located behind the opposite baseline of the tennis court. To measure the ball velocity after forehand

drive impact, a radar gun (SR 3600; Sports-radar, Homosassa, FL, USA) was placed behind the player and in the same location (5 m), height (1 m), and angles (aligned on the top of the net and on the center service line) for both pre- and post-test. The target zone was defined from the baseline and alley line in the forehand side of the opposite court and divided into 4 areas (Figure 1). A ball rebound in the area A1 (1 × 1 m) accounted for one point; in the area A2 (2 × 2 m), for 2 points; in the area A3 (3 × 3 m), for 3 points; in the area A4 (4 × 4 m), for 4 points; and in the court, for 5 points. Another location of the ball rebound resulted in 8 points. The least points were marked, and more accurate were forehand drives. The first set aimed at allowing the players to be familiarized with the ball machine, and only the second set (after 3-minute rest) was used for the subsequent analysis.

Training Protocol

During the 6-week training protocol, the players were instructed to maintain the duration of their tennis training at the same level as such before protocol and not to bring any technical alteration in their forehand drive. Moreover, the RTT was 90 minutes per session, 4 players per court, and conducted by the 2 lead authors. After a general and specific

warm-up (10 minutes), exercises to control the ball direction and the depth in basic strokes (20 minutes) were proposed, followed by exercises for the transition to the net and volley (20 minutes), next exercises for serve and return (15 minutes), and finally tactical games (25 minutes). Free additional practice corresponded to training match. In addition, the participants in HMB and OWR groups included HMB throws or OWR

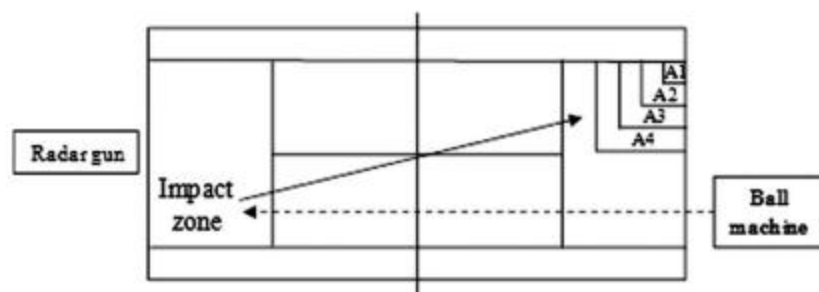


Figure 1. Testing environment, the ball trajectory before impact (dotted arrow), the ball trajectory after impact (full arrow), and A1, A2, A3, and A4 (the 4 areas of the target zone).

TABLE 2. Mean $\pm$ SD of the ball velocity and score before (pre) and after (post) the 6-week training session in the HMB throws group, OWR group, and RTT group.*

		HMB	OWR	RTT
Velocity ($\text{m} \cdot \text{s}^{-1}$)	Pre	32.83 $\pm$ 4.69	29.35 $\pm$ 2.25	31.01 $\pm$ 2.91
	Post	35.5 $\pm$ 3.50	31.39 $\pm$ 2.39	29.62 $\pm$ 3.23
Score (points)	Pre	16.75 $\pm$ 5.29	10.45 $\pm$ 3.20	13.5 $\pm$ 2.68
	Post	18.42 $\pm$ 7.90	11.3 $\pm$ 4.93	12.42 $\pm$ 2.27

*HMB = handled medicine ball; OWR = overweight racket; RTT = regular tennis training.

play, respectively, twice per week for 6 weeks in the precompetitive phase of the yearly periodization. Players in HMB had to perform 6 sets of 6 HMB rotational throws immediately followed by 10 forehand drives with a 1-minute 30-second rest interval between sets. The available HMB mass was 2, 3, or 4 kg (21). Before the first training session, the HMB mass was determined according to the player's ability to perform a throw of 15 m minimal. This pragmatic distance corresponds to an on-court standard, that is, the distance from the baseline to the middle of service box to in the opposite side of the net, and was chosen because of its practical application in a tennis court. The HMB mass remained constant for each player throughout the 6-week training. The players were instructed to throw HMBs at maximal velocity with a similar path than that of the forehand drive and to hit crosscourt forehand drives as fast as possible with a ball rebound in the target zone (Figure 1). The 2 lead authors, who had years of experience in tennis and resistance training with medicine balls, controlled the throwing execution to be as close as possible to the forehand's technique. Players in OWR group had to perform

7–10 sets of 10 crosscourt forehand drives with an OWR with a 1-minute 30-second rest interval between sets. The number of sets increased from 7 to 10 during the 4 first weeks and remained to 10 for the last 2 weeks. The players were instructed to hit the ball as fast as possible with a ball rebound in the target zone (Figure 1). The OWR corresponded to the own racket with 30 g additional mass ($11.15 \pm 3.34\%$).

Each personal racket was customized by adding weights adequately on the racket frame, with no alteration in the racket balance. The OWR was only used during the training session of the protocol.

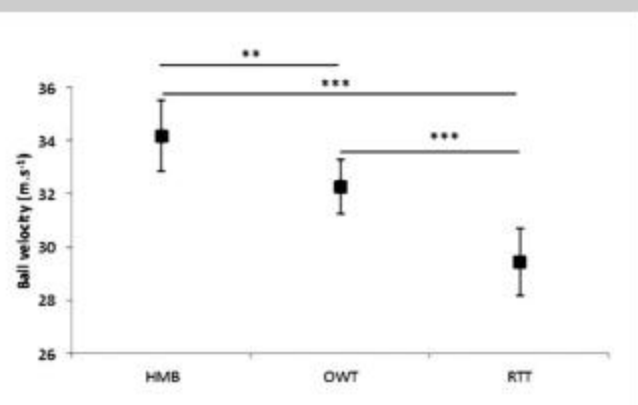
Statistical Analysis

Forehand drive performance was evaluated by the mean velocity of the 5 most accurate forehand drives in the same test-set and by the score of these 5 strokes. Values for all measures are presented as mean values with SD. After verifying the normality and homogeneity of variances, one-way analysis of variance (ANOVA) was performed to compare the characteristics of the 3 groups. In addition, as the pre-test velocity and score were highly correlated to the post-test values, analysis of covariance (ANCOVA) was performed to highlight the differences between groups in post-tests for forehand drive velocity and accuracy. The performance during the pre-test was used as the covariate in each model. When ANCOVA revealed significant difference, the contrast method was applied to test between-group differences. For ANCOVA, F and p values and partial η^2 and its interpretation according to Cohen scale were reported. The level of significance was set at $p \leq 0.05$ for ANOVA and ANCOVA, whereas the Bonferroni's Correction was applied for contrast tests, and then the level of significance was fixed at $p \leq 0.01$. Estimated marginal means with 95% confidence interval from ANCOVA are used in figures, whereas raw means are shown in tables. All statistical analyses were performed on SPSS 11.0 (SPSS, Inc., Chicago, IL, USA).

RESULTS

Analysis of variance revealed no significant difference in the tennis characteristics of the 3 groups of players (Table 1). Table 2 presents the mean $\pm$ SD of the raw ball velocity and score in the three groups before and after training.

Analysis of covariance revealed a significant group effect on the post-training velocity ($F = 14.373$; $p < 0.001$; partial $\eta^2 = 0.42$; large effect). As shown in Figure 2, contrast tests revealed that speed in HMB and OWR groups was significantly higher than in RTT ($p < 0.001$ and $p = 0.001$, respectively). In addition, the post-training speed was

**Figure 2.** Estimated averaged velocity and 95% confidence intervals in the post-test with a pre-test speed estimated at $30.6 \text{ m} \cdot \text{s}^{-1}$ (covariate) for HMB, OWR, and RTT. * shows the significant difference for the contrast tests, with ** for $p = 0.01$ and *** for $p < 0.001$. HMB = handled medicine ball training group; OWR = overweight racket training group; RTT = regular tennis training group.

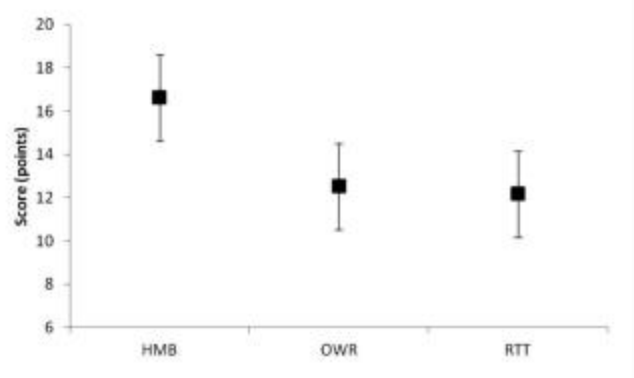


Figure 3. Estimated averaged scores and 95% confidence intervals in the post-test, with pre-test accuracy estimated at 13 points (covariate) for HMB, OWR, and RTT. HMB = handled medicine ball training group; OWR = overweight racket training group; RTT = regular tennis training group.

significantly higher for HMB when compared with OWR ($p = 0.01$).

Concerning the accuracy of the forehand drives (Figure 3), the ANCOVA tended to reveal a significant difference between groups for the score after training ($F = 2.22$; $p = 0.06$; partial $\eta^2 = 0.10$; medium effect). No differences were observed between OWR and RTT groups, but their scores tended to be lower than that for HMB ($p = 0.04$ and $p = 0.02$, respectively).

DISCUSSION

The aim of this study was to examine the effects of 2 training modalities on the forehand drive ball velocity and accuracy. The main results showed that after 6 weeks of training, HMB throws and forehand drives played with OWR conducted to a significant increase in ball velocity when compared with only tennis practice. The effects of HMB throw training were significantly higher than those of forehand drives played with OWR training on ball velocity but tended to deteriorate shot accuracy.

Based on estimated marginal mean values (Figure 2), the increase in ball velocity observed in this study was estimated to about 11% in HMB and 5% in OWR groups (Figure 2). For the latter, the progression was lower with those of previous studies. Indeed, Ravé et al. (16) reported an increase by 12.7% in ball velocity after 7 weeks of dry swing training. This greater difference between pre- and post-training tests can be explained by the longer training protocol and the use of OWR representing 200% of the normal mass to play the dry swings. In baseball, DeRenne et al. (6) showed a stronger progression too, when training including ball hits (+10%), which could be explained by difference in training protocol (baseballs were hitting with alternated overweight, underweight, and standard 30-oz bats for 12 weeks) and initial players' skill level. However, when training conditions were quite similar to those of this study, improvements in swing or

throw speed were ranged between 3 and 12% (5,6). If the comparison of the present results with those of previous works remained difficult, it appeared that, after 6 weeks, significant increase in forehand drive ball speed was observed for the 2 training modalities studied.

Handled medicine ball throws were usefulness to improve tennis forehand drive velocity (Figure 2), as suggested by previous authors (3,18). According with Earp and Kraemer (8), medicine ball throws improve power and velocity in sports, which require a great deal of rotational power, such as tennis and baseball (4,7,20). However, even if the players were instructed to mimic forehand drive path and to throw the HMB as fast as possible, and to play real forehand drives between HMB throws sets, the findings of this study also suggested that this training modalities tended to deteriorate the shot accuracy (Figure 3). The throwing movement was probably done slower than real forehand drive, in relation to a high force component. This increase in force component allowed a great power development at that part of the force-velocity curve but would also generate differences in muscle contraction velocity or muscle coordination or both. As a consequence, HMB throws may be considered as special strength training and may be periodized into the preseason program.

The results of this study also confirmed the efficiency of OWR training, which was lower than that of special strength training (Figure 2). Despite the increase in the workload over the 4 first weeks of the training program, by increasing the number of sets, the total workload in OWR group was lower than in HMB group, whereas this seems to be of importance to design and compare training programs (22). In addition, the longer weekly tennis training reported in OWR than in HMB (Table 1) could also explain the lower ball speed improvement. The increment in the workload through the 6-week program in OWR may be insufficient and then did not optimize the neuromuscular adaptation to the overload stimulus. Interestingly, OWR training allowed the level of shot accuracy to be maintained after the training session, as already observed for tennis forehand drive (16) and swings and pitches (6). This could be explained that the OWR allowed a training stimulus that mimics motions and bioenergetics systems of the tennis activity. As a consequence, OWR training may be considered as specific strength training and may be periodized into the on-season program.

The findings of this study showed that an effect of training transfer occurred when the elements of the supplementary and overloading exercises were similar to those of the primary activity. Even if the methodology used here did not allow the origin of such alterations, caused by the 2 training protocols, to be identified with certainty, it could be hypothesized that they would be mainly nervous and related to the technique, such as inter- and intramuscular coordination and better use of elastic energy. Indeed, this forehand technique evolved strongly with the grips used (more closed), the stance used (more open), and the involvement

of the dominant upper limb segments in the arm internal rotation, which contributes significantly to the production of the racket's speed (12). Elliott et al. (12) estimate the contribution of trunk rotation to 10% and "anatomical rotations" of arm, forearm, and hand to 90% in the racket speed at impact. The kinematic chain of the forehand drive corresponds to a proximal-to-distal sequence (11,19), which is fundamental to generate high racket speed (14). In addition, the forehand technique involves a stretch-shortening cycle (10). The back-swing phase of the forehand drive leads to store energy in stretched muscles and tendons, which is restituted during the muscle concentric contraction of the forward swing phase (10). An optimal use of the stretch-shortening cycle can increase the racket speed by 10–20% (9). Therefore, it could be suggested that this type of specific training may further develop the use of the elastic and neural augmentation that occurs during the stretch-shortening cycle in the forehand drive when special strength training effects could be more related to the rotational speed.

The main limitations of this study were the empirical choice of the baseline distance retained for the HMB throw, and no or few increase in the training load throughout the 6-week program. In addition, despite the precautions to locate the radar on the tennis court, precisely, during the pre- and post-tests, error of measurements could occur, in particular by the deviation between postball impact trajectory and radar directions. Also, this study showed that the 2 training modalities are relevant to improve the forehand drive ball speed and enhance the variety of exercises for physical training in competitive tennis players. Further studies are still needed to optimize the content, duration, and periodization of such training programs. Indeed, it seems to be important to know how the different training methods could be combined in a yearly cycle to maximize the gain of velocity according to the experience of the player. Also, frequent measurement and determination of the optimal loads might be necessary to provide appropriate stimuli to the neuromuscular system and might provide useful information about the effect of training programs and the training status of the player.

To conclude, the main findings of this study demonstrate the efficiency of special strength training, based on HMB throws, and specific strength training, based on OWR forehand drives, to improve the tennis forehand drive ball speed. This study also underlined that only OWR forehand drives maintained the shot accuracy to the contrary of HMB throws, hence suggesting that such exercises for tennis players' strength development may be programmed at different periods.

PRACTICAL APPLICATIONS

Because both HMB throws and OWR training protocols increased forehand drive ball speed in adult tennis players, tennis coaches and physical trainers have a choice of which

protocol to use in their strength training program. For best results, they may periodized the HMB throws into their preseason program, whereas the OWR forehand drives could be included in their on-season program where the goal is to transfer the strength gains in the specific technique of forehand. It must be noted that special and specific trainings will also have their place in an integrated physical preparation whose goal is to improve the determinants of physical condition while practicing the technical activity. This could allow the intermediate tennis players to improve their ball speed without increasing their training duration. Although being quite easy to implement, the presence of a fitness specialist or tennis coach remains necessary to conduct this specific resistance training, to fix the overload and to provide some feedbacks on forehand drive achievement.

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